

Dynamic Programming

Readings:

Azzimonti et al. (2024, ch. 4.4.)

Ljungqvist + Sargent (2018, ch. 2.2, 3, 4)

Stokey, Lucas, Prescott (1989, ch. 4)

The problem

Problem 1

$$\max_{\{y_t, x_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \hat{F}(y_t)$$

subject to

$$x_{t+1} = h(x_t, y_t)$$

$$x_{t+1} \in \Gamma(x_t)$$

$\hat{F}$: instantaneous (flow) objective

y_t : control variables

x_t : state variables

h : transition equation

Γ : feasible set

β : discount factor

Suppose $\Gamma(x_t) = [\underline{x}, \bar{x}(x_t)]$

solve $x_{t+1} = h(x_t, y_t)$ for y_t , so $y_t = \hat{h}(x_t, x_{t+1})$

$$F(x_t, x_{t+1}) = \hat{F}(\hat{h}(x_t, x_{t+1}))$$

Problem 2:

$$\max_{\{x_{t+1} \in \Gamma(x_t)\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t F(x_t, x_{t+1})$$

↑
as in Problem 1

difference: transition eq. has been incorporated into the objective

Suppose F is differentiable in both arguments

Assume interior solution

$$F_2(x_t, x_{t+1}) + \beta F_1(x_{t+1}, x_{t+2}) = 0$$

Transversality condition

$$F_1 = \frac{\partial F}{\partial x_t} \quad F_2 = \frac{\partial F}{\partial x_{t+1}}$$

$$\lim_{T \rightarrow \infty} \beta^T F_1(x_T, x_{T+1}) \cdot x_T = 0$$

$$\left[\frac{\partial F}{\partial x_t} \right] = \frac{\text{utils}}{\text{units } x_t}$$

Example: Neoclassical Growth Model (Planning Problem)

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

s.t.

$$k_{t+1} = k_t(1-\delta) + i_t \quad \leftarrow \text{depreciation rate}$$

$$y_t \geq c_t + i_t$$

$$i_t \leq y_t \leq c_t$$

$$y_t \leq f(k_t)$$

$\leftarrow$ production function

$$y_t = A_t k_t^\alpha$$

$$c_t \geq 0$$

$$k_{t+1} \geq 0$$

k_0 given

for upper bound on k_{t+1}
 $c_t = 0$

$$k_{t+1} + c_t \leq k_t(1-\delta) + f(k_t)$$

if $u' > 0$

$$\begin{aligned} k_{t+1} &= k_t(1-\delta) + f(k_t) - c_t \\ &= h(k_t, c_t) \end{aligned}$$

where generic

$$\begin{array}{c|c} k & x \\ c & y \end{array}$$

$$k_{t+1} \geq 0$$

$$k_{t+1} \leq k_t(1-\delta) + f(k_t)$$

$$\Gamma(k_t) = [0, k_t(1-\delta) + f(k_t)]$$

$$u_{t+1} = u_t \dots$$

$$\Gamma(k_t) = [0, k_t(1-s) + f(k_t)]$$

Problem 1

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

$$k_{t+1} = k_t(1-s) + f(k_t) - c_t$$

$$k_{t+1} \in [0, k_t(1-s) + f(k_t)]$$

Problem 2

$$c_t = k_t(1-s) + f(k_t) - k_{t+1}$$

$$\max_{\{k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(k_t(1-s) + f(k_t) - k_{t+1})$$

$$c_t \in [0, k_t(1-s) + f(k_t)]$$

k_0 given

Dynamic Programming

Define value function as

$$V(x_0) = \max_{\{x_{t+1} \in \Gamma(x_t)\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t F(x_t, x_{t+1})$$

$$V(x_0) \stackrel{(1)}{=} \max_{\{x_{t+1} \in \Gamma(x_t)\}_{t=0}^{\infty}} \left\{ F(x_0, x_1) + \beta \sum_{t=1}^{\infty} \beta^{t-1} F(x_t, x_{t+1}) \right\}$$

assumption

$$\stackrel{\downarrow}{=} \max_{x_1 \in \Gamma(x_0)} \left\{ F(x_0, x_1) + \beta \max_{\{x_{t+1} \in \Gamma(x_t)\}_{t=1}^{\infty}} \left[\sum_{t=1}^{\infty} \beta^{t-1} F(x_t, x_{t+1}) \right] \right\}$$

$$= \max_{x_1 \in \Gamma(x_0)} \left\{ F(x_0, x_1) + \beta \max_{\{x_{t+1} \in \Gamma(x_t)\}_{t=1}^{\infty}} \left[\sum_{t=1}^{\infty} \beta^{t-1} F(x_t, x_{t+1}) \right] \right\}$$

$$= \max_{x_1 \in \Gamma(x_0)} \left\{ F(x_0, x_1) + \beta V(x_1) \right\}$$

↑
only a function of x_1

only a function of x

Features:

- we can maximize "in steps"
- Sequence problem involves choosing $\{x_1, x_2, \dots\}$
- Recursive problem: choose x_i given $V(x_i)$
- Equivalence between sequence and recursive problems "principle of optimality"
- both problems have advantages

Notational change:

replace x_t by x

x_{t+1} by x'

Bellman Equation

$$V(x) = \max_{x' \in \Gamma(x)} \{ F(x, x') + \beta V(x') \}$$

x state

V Value function

functional equation

unknown is the value function

$$V: X \rightarrow \mathbb{R}$$

V only depends on x

Policy Function:

$x' = g(x)$ also only a function of x

$$g(x) = \arg \max_{x' \in \Gamma(x)} \{ F(x, x') + \beta V(x') \}$$

$$V(x) = F(x, g(x)) + \beta V(g(x))$$

The Neoclassical Growth Model

State: k

$$V(k) = \max_{k' \in [0, (1-\delta)k + f(k)]} \{ u((1-\delta)k + f(k) - k') + \beta V(k') \}$$

Properties of the value function

- 1) the value function is often unique
- 2) u u u is often approached by iterations
- 3) If (i) F is strictly increasing in its first argument
(ii) Γ is monotone ($x \leq \tilde{x}$ implies $\Gamma(x) \leq \Gamma(\tilde{x})$)
then V is strictly increasing
- 4) If (i) F is jointly strictly concave in its two arguments
(ii) Γ is convex ($x' \in \Gamma(x)$ and $\tilde{x}' \in \Gamma(\tilde{x})$ then $\forall \theta \in (0,1) \theta x' + (1-\theta)\tilde{x}' \in \Gamma(\theta x + (1-\theta)\tilde{x})$)
then V is strictly concave and the policy function is unique
- 5) The value function is sometimes differentiable and can be used to obtain the Euler equation
- 6) The policy function is sometimes increasing

The Bellman Operator as a contraction

the Bellman Operator

Θ space of continuous, real-valued functions defined on $X \subseteq \mathbb{R}^n$, X closed and bounded interval
 $f \in \Theta : f : X \rightarrow \mathbb{R}$

Define Bellman operator $T : \Theta \rightarrow \Theta$

$$T(f(x)) = \max_{x' \in \Gamma(x)} \{ F(x, x') + \beta \underline{f(x')} \} \quad (*)$$

Some Definitions

Def. 1 : (Metric)

...

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Let M be a set. The function $d: M \times M \rightarrow \mathbb{R}$ is a metric if it satisfies for all $x, y, z \in M$:

1. $d(x, y) = 0$ if and only if $x = y$
2. (Symmetry) $d(x, y) = d(y, x)$
3. (Triangle inequality) $d(x, z) \leq d(x, y) + d(y, z)$

The pair (M, d) is called a metric space

Sup-norm:

$$d(f, g) = \sup_{s \in S} |f(s) - g(s)| \quad \text{for any } f, g \in \Theta$$

The sup-norm is a metric

Metric space (Θ, d)

Definition 2 : (Contraction Mapping)

Let (Θ, d) be a metric space. The mapping $T: \Theta \rightarrow \Theta$ is a contraction on Θ if there exists modulus $\beta \in [0, 1)$ such that

$$d(T(f), T(g)) \leq \beta d(f, g)$$

Two Results

Prop 1 : (Banach's fixed-point theorem)

Let (Θ, d) be a non-empty, complete metric space with contraction mapping $T: \Theta \rightarrow \Theta$ with modulus β . Then

1) T has a unique fixed point f^* , i.e. $T(f^*) = f^*$

2) for any $f_0 \in \Theta$: $d(T(f_0), f^*) \leq \beta d(f_0, f^*)$

Prop 2: (Blackwell's sufficient conditions for a contraction)

Let $T: \Theta \rightarrow \Theta$ be an operator, satisfying

i) monotonicity: If $f, g \in \Theta$ with $f(s) \leq g(s) \forall s$

then $T(f(s)) \leq T(g(s)) \forall s$

ii) discounting: there exists $\beta \in [0, 1)$ such that
 $\forall f \in \Theta$ and for all constant functions k

$$T(f(s) + k) \leq T(f(s)) + \beta k$$

Then T is a contraction with modulus β

Verifying Blackwell's sufficient conditions

Monotonicity: Take $f, g \in \Theta$ with $f(x) \leq g(x) \forall x$

$$\begin{aligned} T(f(x)) &= \max_{x' \in F(x)} \{ F(x, x') + \beta f(x') \} \\ &\leq \max_{x' \in F(x)} \{ F(x, x') + \beta g(x') \} \\ &\stackrel{\text{def}}{=} T(g(x)) \end{aligned}$$



Discounting:

Take $f \in \Theta$, $k \in \Theta$, k constant

$$\begin{aligned} T(f(x) + k) &= \max_{x' \in F(x)} \{ F(x, x') + \beta (f(x') + k) \} \\ &= \max_{x' \in F(x)} \{ F(x, x') + \beta f(x') + \beta k \} \\ &= T(f(x)) + \beta k \end{aligned}$$

✓ for $\beta < 1$

Hence: The Bellman operator is a contraction

Note: Set of bounded and real-valued functions

Note: Set of bounded and real-valued functions is complete, so (B, d) is a complete metric space

$\Rightarrow$ Banach's fixed point theorem applies

1) The Bellman operator T has a unique fixed point: $T(V) = V$

Hence, we can write:

$$V(x) = \max_{x' \in \Gamma(x)} \{ F(x, x') + \beta V(x') \}$$

2) Value function iteration

$\rightarrow$ the fixed point is approached by iterations

initial guess V_0

$$\text{then } d(T(V_0), V) \leq \beta d(V_0, V)$$

$$\text{Hence: } d(T^u(V_0), V) \leq \beta d(T^{u-1}(V_0), V) \leq \dots \leq \beta^u d(V_0, V)$$

$$\lim_{u \rightarrow \infty} T^u(V_0) = V$$

Algorithm:

1) Set V_0 to initial guess, eg. $V_0(x) = 0 \quad \forall x$

2) Iterate for $u = 0, 1, \dots$ using the following recursion:

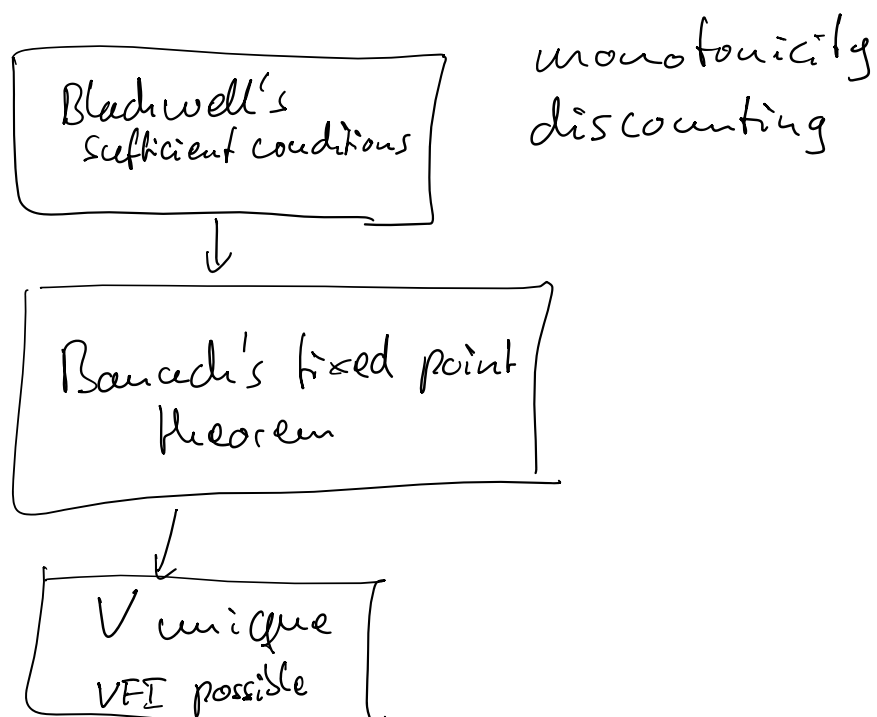
$$V_{u+1}(x) = \max_{x' \in \Gamma(x)} \{ F(x, x') + \beta V_u(x') \}$$

Remark:

sometimes F is not bounded in economic applications,

Remark:

oftentimes F is not bounded in economic applications,
 e.g. $f(k) = k^\alpha$ on $[0, \infty)$. This implies that the set
 Θ on which the Bellman Operator is defined also
 contains unbounded functions. Then (Θ, d) is not
 generally complete and hence Banach's fixed
 point theorem may not apply...



Obtaining the Euler Equation

$$V(x) = \max_{x' \in F(x)} \{ F(x, x') + \beta V(x') \}$$

Assume differentiability, interior solution

$$F_2(x, x') + \beta \underline{V'(x')} = 0 \quad (*)$$

policy $x' = g(x)$ solves this equation

$$-1 \quad -f_2(x, g(x)) \quad -\beta V'(g(x)) = 0$$

only $x = g(x)$ solves this eq.

$$F_2(x, g(x)) + \beta V'(g(x)) = 0$$

$$V(x) = F(x, g(x)) + \beta V(g(x))$$

$$\begin{aligned} V'(x) &= F_1(x, g(x)) + F_2(x, g(x)) g'(x) + \beta V'(g(x)) \cdot g'(x) \\ &= F_1(x, g(x)) + \underbrace{(F_2 + \beta V')}_{=0(x)} g'(x) \end{aligned}$$

Shift 1 period ahead

$$\begin{aligned} V'(x') &= F_1(g(x), g(g(x))) \\ &= F_1(x', g(x')) \end{aligned}$$

Functional Euler equation

$$F_2(x, g(x)) + \beta F_1(g(x), g(g(x))) = 0$$

compare this to our Euler equation from the sequence problem

$$F_2(x_t, x_{t+1}) + \beta F_1(x_{t+1}, x_{t+2}) = 0$$

Example: A stochastic consumption-saving problem

Hlt:

$$\max_{\{c_t, a_{t+1}\}_{t=0}^{\infty}} E_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$$

s.t.

$$c_t + \frac{a_{t+1}}{1+r} = y_t + a_t$$

Assumption
 r is constant

$$a_{t+1} = s_t(1+r)$$

uncertainty about y_t

plus pg condition

y_t follows a 2-state Markov process

y_L low

$$\frac{P(y_L | y_L)}{P(y_L | y_H)}$$

$$P(y_H | y_L) = 1 - P(y_L | y_L)$$

y_H high

$$\frac{P(y_H | y_H)}{P(y_H | y_L)}$$

$$P(y_L | y_H) = 1 - P(y_H | y_H)$$

1) State variables

a, y

2) Bellman eq.

$$V(a, y) = \max_{a'} \left\{ u\left(a + y - \frac{a'}{1+r}\right) + \beta E[V(a', y') | y] \right\}$$

$$E[V(a', y') | y] = P(y_H | y) \cdot V(a', y_H) + P(y_L | y) \cdot V(a', y_L)$$

FOC

$$0 = u'(c) \cdot \left(-\frac{1}{1+r}\right) + \beta E\left[\frac{\partial V(a', y')}{\partial a'} | y\right]$$

Suppose $a' = g(a, y)$ policy fn.

$$V(a, y) = u\left(a + y - \frac{g(a, y)}{1+r}\right) + \beta E[V(g(a, y), y') | y]$$

$$V(a, y) = u\left(a + y - \frac{g(a, y)}{1+r}\right) + \beta E[V(g(a, y), y') | y]$$

$$\begin{aligned} \frac{\partial V(a, y)}{\partial a} &= u'(c) - \frac{1}{1+r} u'(c) \cdot \frac{\partial g(a, y)}{\partial a} + \beta E\left[\frac{\partial V(g(a, y), y')}{\partial a'} \cdot \frac{\partial g(a, y)}{\partial a} \middle| y \right] \\ &= u'(c) + \underbrace{\left[-\frac{u'(c)}{1+r} + \beta E\left[\frac{\partial V(g(a, y), y')}{\partial a'} \middle| y \right] \right]}_{=0 \text{ by FOC}} \frac{\partial g(a, y)}{\partial a} \end{aligned}$$

$$= u'(c)$$

$$\frac{\partial V(a', y')}{\partial a'} = u'(c')$$

$\Rightarrow$ Euler equation

$$u'(c) = \beta(1+r) E[u'(c') | y]$$

$$\begin{aligned} c &= a + y - \frac{a'}{1+r} \\ &= a + y - \frac{g(a, y)}{1+r} =: \underline{\underline{h(a, y)}} \end{aligned}$$

$$E[u'(c') | y] = E[u'(h(a', y')) | y]$$

$$= \sum_{y' \in \{y_L, y_H\}} p(y' | y) u'(h(a', y'))$$